

Literature Review: Banking CECL Regulation in Accounting and Finance (2021–2026)

Prepared 21 September 2026 · Structured literature review with Chain-of-Verification citation checks · 60+ sources (published articles, SSRN and central-bank working papers, regulatory documents)

Date: 2026-09-21 **Query:** “Conduct a literature review on banking CECL regulation in accounting and finance, published in the last five years” (follow-up: “search SSRN for working papers”) **Window:** September 2021 – September 2026, plus three early-2021 anchor papers that the later literature builds on (Beatty & Liao; Wheeler; López-Espinosa, Ormazabal & Sakasai). **Sources searched:** web search (general + restricted to ssrn.com), Federal Reserve / ECB / IASB research pages. Direct fetches of publisher pages (Wiley, AAA, Elsevier, SSRN) are blocked in this environment, so bibliographic details come from search-result snippets and were cross-checked in a fresh-context verification pass (see Post-Flight block at the end). Two entries (the JAPP 2025 management-forecasts article and the Research Square mortgage preprint) still lack author names; resolve with library access before citing.

Summary

The Current Expected Credit Loss (CECL) standard (ASU 2016-13, ASC 326) replaced the incurred-loss (IL) model for U.S. GAAP filers: large SEC filers adopted on 1 January 2020, and all remaining banks on 1 January 2023. Because the effective date coincided with the COVID-19 shock, and because the CARES Act made 2020 adoption optional for a subset of banks, the last five years produced an unusually clean set of natural experiments. The literature has moved quickly from pre-adoption simulation and theory (2018–2020) to post-adoption empirical work, and by 2024–2026 the core findings are appearing in the top accounting journals (*The Accounting Review*, *Journal of Accounting Research*, *Journal of Accounting and Economics*, *Contemporary Accounting Research*).

Three findings are now well established. **First, CECL made provisions more forward-looking and timelier.** Kim, Kim, Kleyменова & Li (TAR 2026) show CECL banks’ provisions better reflect future local economic conditions and that CECL banks experience fewer subsequent defaults; the Federal Reserve’s own pandemic post-mortem (Loudis et al. 2021) documents that adopters’ allowances rose sharply and early in 2020. **Second, CECL has real effects on credit supply, concentrated in capital-constrained banks and in stress periods.** Chen, Dou, Ryan & Zou (TAR 2025) find adopters cut loan growth more during the COVID recession, especially low-capital banks; Yang (TAR 2025) finds capital-constrained banks slowed total and residential loan growth after CECL’s approval; Granja & Nagel (JAE 2025) find modest consumer-loan rate increases, larger for weakly capitalized banks and disadvantaged borrowers; Hu (SSRN 2024) finds reduced small-business originations in distant markets. **Third, the procyclicality question remains genuinely contested.** The theory (Mahieux, Sapra & Zhang, JAR 2023) says timely provisioning enables earlier intervention but changes ex ante loan origination in either direction, and that capital regulation should be looser under CECL only when early loss estimates are precise enough or risk-shifting incentives are mild; the COVID evidence is split between “CECL amplified the credit contraction” (Chen et al. 2025; Chen & Huang, JAR 2026 for China’s ECL adopters) and “CECL reduced lending’s sensitivity to the cycle” (Chen, Lin & Yang, SSRN 2023, on mortgage approvals and house prices; Loudis et al. 2021 finding no clear lending effect).

A second strand concerns **information and discretion**. Evidence on whether CECL improved the information environment is mixed: Gee, Neilson, Schmidt & Xie (SSRN 2022, revised) find CECL allowances are

decision-useful for both small and large banks, and Wheeler (JAR 2021) shows investors priced unrecognized expected losses even before adoption; but Bonsall, Schmidt & Xie (SSRN 2025) and Bonaldi, Liang & Yang (SSRN 2023) find analyst forecast properties and earnings-announcement informativeness deteriorated, and Wheeler (SSRN 2025) finds CECL raised the macroeconomic content of provisions while lowering bank-specific credit-loss content. Discretion has not disappeared: European IFRS 9 evidence (Behn & Couaillier, JAE 2026; Novotny-Farkas, Oberson & Renner, SSRN 2024) shows capital-constrained banks provision less for the same exposures, and U.S. community-bank evidence (He, Mamun & Mishra, BAR 2025) shows income smoothing persists. A newer sub-strand studies forecast quality itself: Bhojraj, Kleymenova, Liu, Lu & Sengupta (JAE forthcoming) document overreaction in banks' supervisory loss forecasts, and Canals-Cerdá (Philadelphia Fed 2024, 2026) shows CECL is mechanically more sensitive to macro-forecast error than IL.

The most active frontiers as of 2026 are (i) distributional effects of expected-loss provisioning on household and SME credit access, (ii) the interaction between CECL allowances and the supervisory stress test / stress capital buffer, which the Federal Reserve has explicitly deferred, and (iii) the 2023 adoption wave at community banks, which offers a second, non-pandemic natural experiment that is still largely unexploited.

Key Papers

Ordered roughly by relevance to “CECL regulation and banks.” Publication status noted (published / forthcoming / working paper).

Mahieux, Sapra & Zhang (2023) — CECL: Timely Loan Loss Provisioning and Bank Regulation

- **Status:** Published, *Journal of Accounting Research* 61(1): 3–46.
- **Main contribution:** The canonical theory paper. Models the trade-off between accuracy and timeliness in loss recognition and asks how CECL/IFRS 9 should interact with capital regulation.
- **Method:** Analytical model of a bank with asset-substitution incentives and a regulator who can intervene.
- **Key finding:** Timely provisioning improves efficiency by allowing earlier intervention against ex post asset substitution, but banks anticipate that intervention and originate *either safer or riskier* loans ex ante depending on the regime. Regulatory capital under CECL should be looser than under incurred loss only when early loss estimates are sufficiently precise and/or risk-shifting incentives are not too severe; the result is conditional, not a blanket call for lower capital.
- **Relevance:** Provides the benchmark predictions (capital-constrained banks react most; ex ante origination-incentive channel) that the empirical papers below test.

Kim, Kim, Kleymenova & Li (2026) — CECL Standard and Banks' Information Production

- **Status:** Published, *The Accounting Review* 101(3): 493–526. Earlier versions: FEDS Working Paper 2023-063; SSRN 4117869 / 4601190.
- **Main contribution:** First systematic evidence that CECL changed what banks *know*, not only what they report.
- **Method:** Difference-in-differences comparing 2020 adopters to non-adopters (CARES Act deferral), with local-economic-conditions data and textual analysis of provision disclosures.

- **Key finding:** CECL banks' provisions become timelier and better reflect future local economic conditions; disclosures become longer, more forward-looking and more quantitative; CECL banks experience lower loan default rates. Effects are stronger for banks investing more in systems and human capital.
- **Relevance:** The strongest “benefit” result in the literature; the counterweight to the real-effects cost literature.

Chen, Dou, Ryan & Zou (2025) — CECL and Banks' Lending during Stress Periods: Evidence from the COVID-19 Recession

- **Status:** Published, *The Accounting Review* 100(1): 113–138. SSRN 4110109 (earlier title: “Does the Current Expected Credit Loss Approach Decrease the Procyclicality of Banks' Lending?”).
- **Main contribution:** Direct test of the procyclicality concern using the 2020 adoption / CARES deferral split.
- **Method:** DiD on quarterly Call Report data; heterogeneity by regulatory capital and loan impairment; county-level real outcomes.
- **Key finding:** Adopters increased provisions and reduced loan growth more than non-adopters during the recession. The contraction is stronger for adopters with low regulatory capital and low impairment, and concentrated in commercial loans. Counties with higher CECL-adopter market share saw larger unemployment increases and slower recoveries.
- **Relevance:** The leading “CECL is procyclical in a downturn” result; anchors the open debate.

Yang (2025) — How Does Loan Loss Accounting Influence Bank Lending? Evidence from CECL

- **Status:** Published, *The Accounting Review* 100(1): 465–490 (DOI 10.2308/TAR-2022-0511).
- **Main contribution:** Isolates the *capital-cost* channel of CECL from the pandemic by using the standard's approval (2016), not its effective date.
- **Method:** DiD around CECL's approval, exploiting cross-sectional variation in capital constraints.
- **Key finding:** Capital-constrained banks reduce growth of total loans and residential loans after approval. One search-result abstract also reports that they sell more of the residential loans they originate shortly after origination (originate-to-distribute); this secondary result was not independently corroborated in verification.
- **Relevance:** Shows lending effects pre-date and are independent of COVID; the securitization margin is a novel adjustment channel.

Granja & Nagel (2025) — Current Expected Credit Losses and Consumer Loans

- **Status:** Published, *Journal of Accounting and Economics* 80(1): 101781. SSRN 4324118.
- **Main contribution:** Loan-level evidence on pricing and quantity effects of CECL in consumer credit.
- **Method:** TransUnion credit-bureau data covering millions of consumer loans; adopter vs. non-adopter design with borrower-level heterogeneity.
- **Key finding:** Higher reserve requirements after CECL induce a statistically significant but economically moderate increase in loan interest rates; effects are larger for weakly capitalized banks and larger still for disadvantaged borrowers at weakly capitalized banks.
- **Relevance:** Best-identified pricing evidence; introduces the distributional dimension.

Aaron, Jia, Ng & Zhang (2025) — Current Expected Credit Loss Model Adoption

- **Status:** Published, *Contemporary Accounting Research* 42(4): 2915–2948 (DOI 10.1111/1911-3846.13078).
- **Main contribution:** Treats the 2020 adoption decision itself as the outcome, addressing the selection problem in the DiD literature above.
- **Method:** Cross-section of the adoption choice under the CARES Act option, then panel analysis of pandemic-era provisioning.
- **Key finding:** Banks more exposed to the pandemic were *less* likely to adopt CECL in 2020, more so when they had more lending opportunities, more competition, or worse loan quality. Adopters reported higher provisions early in the pandemic and lower or negative provisions later.
- **Relevance:** Essential caveat for every 2020-adopter DiD: adoption was endogenous to pandemic exposure.

Ertan (2025) — Expected Losses, Unexpected Costs? Evidence from SME Credit Access under IFRS 9

- **Status:** Published, *The Accounting Review* 100(4): 443–473. SSRN 3504708.
- **Main contribution:** The cleanest evidence on expected-loss provisioning and SME credit, using Europe's IFRS 9 transition.
- **Method:** Bank-, borrower- and loan-contract-level DiD around IFRS 9 adoption.
- **Key finding:** Affected banks cut SME lending by more than 10%; rejection rates, spreads and collateral requirements rose; loan amounts, maturities and borrower investment fell. Effects persisted into COVID.
- **Relevance:** Complements Hu (2024) on U.S. small-business lending; suggests opaque, bank-dependent borrowers bear the cost of forward-looking provisioning.

Chen & Huang (2026) — A Tale of Two Banks: When Credit Loss Models Meet Economic Crises

- **Status:** Published, *Journal of Accounting Research* (DOI 10.1111/1475-679X.70030). SSRN 5719602.
- **Main contribution:** Administrative loan-level and firm-level evidence from China, where ECL and ICL banks coexisted at the onset of COVID.
- **Key finding:** ECL banks reduced credit supply and became more prudent after the pandemic began relative to ICL banks.
- **Relevance:** External validity for the Chen et al. (2025) U.S. result; shows the procyclicality pattern is not a U.S. artifact.

Behn & Couaillier (2026) — Weak Capital, Weak Provisions: Credit Risk Provisioning under IFRS 9

- **Status:** Published, *Journal of Accounting and Economics* 81(3). ECB Working Paper 2841 (“Same same but different”); SSRN 4545496.
- **Method:** Euro-area credit-register (AnaCredit-type) data allowing within-borrower, within-period comparison across banks.
- **Key finding:** Less-capitalized banks provision less than better-capitalized peers *for the same loan to the same borrower in the same period*. Most provisioning still occurs at default; aggregate provisioning

patterns are not fundamentally changed by IFRS 9.

- **Relevance:** Strongest evidence that expected-loss discretion is used for capital management; directly relevant to the U.S. debate on CECL model overlays.

Qiang & Wang (2024) — CECL and Conditional Conservatism of Banks, with Spillover to Borrowers

- **Status:** Published, *The Accounting Review* 99(6): 389–420. SSRN 4868246.
- **Key finding:** Under CECL banks increase conservatism in loan losses but decrease conservatism in other earnings components; their borrowers increase conditional conservatism by about 40%, roughly twice the banks' own increase.
- **Relevance:** Introduces the borrower-side reporting spillover; supports the “banks monitor harder under ECL” mechanism also found by Li, Ng & Saffar (Management Science 2024) for IFRS 9.

Lu & Nikolaev (2022) — Expected Loan Loss Provisioning: An Empirical Model

- **Status:** Published, *The Accounting Review* 97(7): 319–346. SSRN 3344657.
- **Main contribution:** A benchmark model for lifetime expected losses.
- **Method:** Cross-sectional loss prediction combined with a high-dimensional dynamic factor model of aggregate losses over the cycle; out-of-sample evaluation.
- **Key finding:** Predicts long-horizon losses substantially more accurately than Harris, Khan & Nissim (2018) and other alternatives, and detects bank failures better.
- **Relevance:** Useful as an external benchmark for “how much of reported CECL allowance is predictable”; feeds the Wheeler (2025) informativeness question.

Bhojraj, Kleymenova, Liu, Lu & Sengupta (2025) — Imperfect Expectations in Loan Loss Forecasts

- **Status:** Accepted, *Journal of Accounting and Economics* (per authors' site); SSRN 4858012 (June 2024); presented at Boston Fed stress-testing conference 2025.
- **Method:** Confidential FR Y-14A supervisory forecasts of net charge-offs; tests of forecast-error predictability.
- **Key finding:** Revisions of charge-off forecasts are negatively associated with subsequent forecast errors (overreaction, consistent with the representativeness heuristic), stronger at longer horizons and for banks with longer-maturity loans; overreaction also appears in banks' regional macro forecasts.
- **Relevance:** If banks overreact, CECL (which loads on long-horizon forecasts) will be *more* procyclical than rational-expectations models predict.

Beatty & Liao (2021) — What Do Analysts' Provision Forecasts Tell Us about Expected Credit Loss Recognition?

- **Status:** Published, *The Accounting Review* 96(1): 1–21.
- **Key finding:** Controlling for recognized provisions, consensus analyst provision forecasts are incrementally associated with future NPLs and returns, implying the IL regime left decision-useful expected-loss information unrecognized.

- **Relevance:** Pre-adoption anchor for the “CECL adds information” hypothesis.

Wheeler (2021) — Unrecognized Expected Credit Losses and Bank Share Prices

- **Status:** Published, *Journal of Accounting Research* 59(3): 805–866.
- **Key finding:** A vintage-analysis measure of lifetime expected losses is negatively priced by the market, more strongly for large banks; recorded allowances were on average below estimated expected losses, foreshadowing capital effects on adoption.
- **Relevance:** Together with Beatty & Liao, establishes what investors could infer pre-CECL. Wheeler’s 2025 SSRN follow-up (“Assessing CECL”) finds CECL raised macro content but lowered bank-specific content of provisions.

López-Espinosa, Ormazabal & Sakasai (2021) — Switching from Incurred to Expected Loan Loss Provisioning: Early Evidence

- **Status:** Published, *Journal of Accounting Research* 59(3): 757–804.
- **Relevance:** The first large-sample IFRS 9 transition paper; the European anchor for the ECL-vs-ICL comparison.

Kim (2022) — Delays in Banks’ Loan Loss Provisioning and Economic Downturns: Evidence from the U.S. Housing Market

- **Status:** Published, *Journal of Accounting Research* 60(3): 711–754.
- **Relevance:** Quantifies the cost of the IL regime’s delayed recognition in the 2007–09 housing bust, motivating CECL.

Loudis, Pechenik, Ranish, Vojtech & Xu (2021) — New Accounting Framework Faces Its First Test: CECL During the Pandemic

- **Status:** FEDS Notes, December 2021.
- **Key finding:** Day-one adoption raised allowances ~37%; excluding adoption effects, allowances rose ~76% in H1 2020 vs. 2019Q4. Adopters’ provisioning was more responsive to the outlook, with wide variation by loan type; limited evidence that CECL reduced lending in the pandemic, with “other consumer” loans the only category showing a statistically significant relative decline and no association with C&I, CRE, or residential real estate loan growth. (The frequently cited “consumer loan growth ~3 pp lower at CECL banks, driven by credit cards” result is from the Bank Policy Institute’s separate study of the same episode, not from this note.)
- **Relevance:** The regulator’s own read; note the contrast with Chen et al. (2025) on commercial lending.

Hu (2024) — The Real Effects of CECL on Banks’ Small Business Lending

- **Status:** Working paper, SSRN 4934238 (August 2024).
- **Key finding:** Adopters cut small-business originations in counties far from their branches, concentrated in loans under \$100k; no such pattern for purchased loans or lien mortgages. Interpreted as CECL raising information-production costs for opaque distant borrowers.

Canals-Cerdá (2024, 2026) — CECL Implementation and Model Risk (Philadelphia Fed WP 24-03; superseded by WP 26-09)

- **Status:** Working papers, Federal Reserve Bank of Philadelphia.
- **Key finding:** CECL is mechanically more sensitive to macro-forecast and model-specification error than IL; a flexible machine-learning modelling framework can be estimated quickly on tens of millions of consumer loans without forecasting-performance penalty.
- **Relevance:** The model-risk literature; relevant to the “reasonable and supportable forecast” debate.

Novotny-Farkas, Oberson & Renner (2024) — IFRS 9 under Stress

- **Status:** Working paper, SSRN 4918921 (August 2024).
- **Key finding:** European quarterly data 2018–2022 show income smoothing, delayed recognition and subjective post-model adjustments; provisions on unimpaired assets are procyclical but prudential filters mitigated lending procyclicality in COVID; under-provisioning on impaired assets is concentrated in weakly capitalized and COVID-exposed banks and associated with reduced lending.

Thematic Organization

Theoretical Contributions

See the Theory Deep-Dive section below for a paper-by-paper treatment.

- **Timeliness vs. accuracy and the capital-regulation interaction** (Mahieux, Sapra & Zhang 2023). Timely recognition is efficient when it enables intervention, but it changes ex ante origination incentives in a regime-dependent direction; capital requirements should fall with timelier provisioning only when early loss estimates are precise or risk-shifting incentives are mild. Related earlier models (Abad & Suarez 2018; Chae et al. 2018) are pre-window but remain the reference for procyclicality simulations.
- **Forecast imperfection** (Vidinova 2023, SSRN 4628656, “Forward-Looking Loan Loss Provisioning under Imperfect Forecasts”; Bhojraj et al. 2025). If forecasts are noisy or biased, forward-looking provisioning can be *more* procyclical than incurred-loss provisioning.
- **Structural quantification** (Huber 2022, SSRN 4120628; listed as R&R at JAR on the author’s site). An estimated bank model reports that CECL raises reserves and reduces lending; two circulating versions give different magnitudes (14.7% reserves / 5.16% lending in one, 16% / 3.15% in another), so cite the dated SSRN version explicitly.
- **Evergreening** (Eufinger, Ramirez-Chiang & Sakasai 2026, SSRN 7216299). ECL weakens banks’ incentive to roll over loans to non-viable “zombie” firms because forward-looking provisions must be booked regardless; empirically, earlier IFRS 9 implementers cut lending growth to zombies.

Empirical Findings

(a) Provisioning behaviour and timeliness. Consensus that CECL/IFRS 9 provisions respond faster to the outlook (Loudis et al. 2021; Kim et al. 2026; Elashmawy & Kallunki 2026 for Europe, who also find IFRS 9 provisions predict fewer NPL adverse events and lower charge-offs). Volatility rose: Jia, Kanagaretnam & Zhang (JAAF 2025) report higher provision and allowance volatility for adopters, alongside timelier and more

valid provisions and higher earnings response coefficients (the specific magnitudes quoted in some summaries, about half a standard deviation and a doubling of volatility, were not verifiable here); Bonaldi, Liang & Yang (2023) show adopters were more likely to book negative provisions in late 2020.

(b) Discretion did not go away. Behn & Couaillier (2026) and Novotny-Farkas et al. (2024) in Europe; He, Mamun & Mishra (BAR 2025) show U.S. community banks smooth income via provisions and shifted toward expected-loss behaviour before formal 2023 adoption; the Kansas City Fed (2023) documents a modest average day-one allowance increase of 3.76% at community banks in 2023, larger for \$1–10bn banks (7.53%) than for sub-\$1bn banks (3.08%), attributed to smaller banks' heavier pre-CECL use of qualitative factors.

(c) Real effects on credit. Contraction in stress (Chen et al. 2025; Chen & Huang 2026), capital-cost channel independent of stress (Yang 2025), pricing (Granja & Nagel 2025), SME/small-business access (Ertan 2025; Hu 2024), mortgage allocation (Research Square 2026 preprint linking post-CECL provision pressure to larger Black–White denial gaps in home-improvement and refinance channels at banks over \$10bn; author not identified in search results, treat as preliminary). Counter-evidence: Chen, Lin & Yang (2023) find mortgage approval rates became *less* sensitive to house-price growth after CECL (their secondary claim of dampened local housing booms was not independently corroborated); Loudis et al. (2021) find no C&I/CRE/RRE effect. Asymmetry: Buchetti, Perdichizzi & Santoni (Economics Letters 2025) show lending contracts immediately on IFRS 9 stage downgrades but recovers only slowly on upgrades.

(d) Information environment and capital markets. Positive: Gee et al. (2022, revised) — CECL allowances decision-useful for small and large banks, new information mainly for small banks; Kim et al. (2026) — richer disclosure; JAPP 2025 (“Expected loss recognition and banks’ management forecasts”) — banks issue more, and more precise, management forecasts after ELM adoption worldwide; Hu (2024b) — negative information spillovers from bellwether banks’ provisions to non-announcing banks and constrained non-financial firms shrink after CECL. Negative: Bonsall, Schmidt & Xie (2022, 2025) — analyst provision forecasts less accurate, more dispersed, lower coverage; earnings-announcement informativeness falls most where CECL raises provision volatility and where sophisticated investors are present; Bable, Wong & Wynes (2023) — interviewed users perceive lower decision usefulness and comparability; Wheeler (2025) — more macro, less bank-specific information.

(e) Spillovers to borrowers. Qiang & Wang (2024) on borrower conservatism; Li, Ng & Saffar (Management Science 2024) show IFRS 9-driven monitoring shifts firms from bank to public debt; Guo, Jia, Ng & Zhu (SSRN 2025) on cross-border borrowing.

Methodological Innovations

- **CARES Act deferral as an instrument for adoption timing** (Chen et al. 2025; Kim et al. 2026; Bonaldi et al. 2023; Jia et al. 2025), with Aaron et al. (2025) showing the choice is endogenous to pandemic exposure — later papers should model selection or use approval-date designs (Yang 2025).
- **Credit-register within-borrower designs** (Behn & Couaillier 2026; Chen & Huang 2026; Ertan 2025) that hold borrower demand fixed.
- **Credit-bureau loan-level pricing data** (Granja & Nagel 2025) and **HMDA application-level data** (Chen, Lin & Yang 2023; Research Square 2026).
- **Supervisory forecast data (FR Y-14A)** for testing forecast rationality (Bhojraj et al. 2025).
- **Textual measures of forward-looking disclosure** (Kim et al. 2026).
- **Out-of-sample lifetime-loss benchmarks** (Lu & Nikolaev 2022) and ML model-risk frameworks (Canals-Cerdá 2024/2026).

Open Debates

1. **Is CECL procyclical?** Theory says “depends on capital regulation”; U.S. COVID evidence says yes for commercial lending at low-capital banks (Chen et al. 2025) but no for mortgage approvals (Chen, Lin & Yang 2023) and no aggregate effect per the Fed (Loudis et al. 2021). COVID’s fiscal support and regulatory relief (capital transition, PPP) confound all of these.
 2. **Did information quality rise or fall?** Bank-level information production improved (Kim et al.), but external users (analysts, interviewees) report worse predictability; Wheeler (2025) reconciles part of this as a macro-vs-idiosyncratic composition shift.
 3. **How much discretion remains, and is it harmful?** European evidence says capital management survives ECL; U.S. evidence on post-model overlays (“Q-factors”) under CECL is thin.
 4. **Should CECL enter the supervisory stress test?** The Federal Reserve’s 2024–2025 stress-test proposals explicitly do *not* incorporate CECL, citing that the current allowance framework already projects four quarters ahead; industry analysis (BPI) argues incorporation would raise capital requirements via larger peak-to-trough declines.
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Theory Deep-Dive: What the Analytical Literature Says

The theory literature on expected-loss provisioning is small (about ten papers) but has converged on a shared framework: provisioning rules matter because regulatory capital is computed from accounting numbers, so the timing and precision of loss recognition change *when* a regulator can intervene and *how* a bank chooses loans ex ante. Two families of models exist. Analytical accounting-theory models (Mahieux, Sapra & Zhang; Bertomeu, Mahieux & Sapra; Fleer; Lu & Zhang; Vidinova) isolate mechanisms in two- or three-period settings. Dynamic quantitative models (Goncharenko & Rauf; Abad, Ikeda & Suarez; Huber; Buesa, Población & Tarancón; Kund & Rugilo) calibrate or estimate the size of the procyclicality effect. The two families reach compatible conclusions, summarized after the papers.

The papers

Mahieux, Sapra & Zhang (JAR 2023) — the benchmark model. A bank originates loans whose risk it can shift after origination; a regulator intervenes when accounting information reveals trouble. The incurred-loss (IL) model reports late but accurately; the expected-loss (EL) model reports early but noisily (the accuracy-versus-timeliness trade-off). Results: (i) EL prompts earlier corrective action in bad times, which curbs ex post asset substitution; (ii) anticipating that intervention, the bank changes its ex ante origination, and the direction is not fixed: EL can induce *either safer or riskier* loans; (iii) EL raises efficiency when banks are insufficiently capitalized or when intervention is likely to be effective, but *impairs* efficiency when banks are moderately capitalized and intervention is sufficiently costly; (iv) the regulator can restore efficiency by tailoring capital to loss information, and capital under CECL should be looser than under IL when early loss estimates are sufficiently precise and/or risk-shifting incentives are not too severe. The policy message is that CECL and capital regulation must be designed jointly.

Bertomeu, Mahieux & Sapra (TAR 2023) — accounting measurement and capital requirements as joint instruments. The regulator cannot commit to ex ante efficient intervention and reacts to ex post accounting reports. Considered separately, capital requirements and accounting precision are substitutes (either can restrain risk). Considered jointly, they are complements: better expected-loss information lets the regulator set looser capital requirements and still achieve efficient credit decisions, which spurs lending. This is the formal basis for

the claim, repeated in the CECL debate, that regulators should not simply layer CECL on top of unchanged Basel capital rules.

Fleer (JAPP 2025) — optimal recognition thresholds. Models a bank that both supplies credit and provides liquidity to depositors. Recognizing significant lifetime expected losses disciplines ex ante risk-taking but can distort ex post liquidity provision (a loss report can trigger withdrawals). A unique conditionally optimal recognition threshold balances the two, and it depends on the bank's environment. The paper reads this as support for IFRS 9's principle-based "significant increase in credit risk" (SICR) trigger over a one-size-fits-all rule, and by implication as a caution against CECL's day-one lifetime recognition for every exposure.

Lu & Zhang (Advances in Accounting 2026) — time-varying capital ratios under CECL. Chooses the capital ratio that maximizes the joint surplus of debtholders and equityholders when the regulator observes CECL reports. The optimal ratio depends on two cycle parameters: persistence of shocks to loan quality (asset side) and intertemporal change in deposit liquidity demand (liability side). For highly levered banks the optimal initial ratio is tighter when a cycle turn is anticipated and looser when the current phase is expected to persist (the working-paper version phrases this as procyclical under rosy prospects and countercyclical under gloomy ones). The point for policy is that a static ratio is never optimal once provisions are forward-looking.

Vidinova (working paper, 2023) — CECL under behavioral forecasts. Builds a model of provisions, lending, output and stability under CECL, first with rational expectations and then with forecasts shaped by the representativeness heuristic (overreaction to recent news). Overreaction produces under-provisioning, excess lending and risk-taking after good news and the reverse after bad news, so CECL becomes procyclical precisely because it loads on forecasts that cannot see cycle turns. The optimal minimum capital requirement is time-varying and tighter when default risk is high; with behavioral forecasts the constraint must bind even when equity is high and, holding true risk fixed, the optimal requirement falls in the bank's *perceived* risk (because over-pessimistic banks already over-provision). Bhojraj et al. (JAE forthcoming) supply the empirical counterpart: supervisory loss forecasts do overreact.

Goncharenko & Rauf (IJCB 2024) — "still too much, too late." A dynamic model of a bank facing minimum capital regulation and taxes, where provisioning requirements change intertemporal trade-offs. If the bank cannot anticipate the downturn, it recognizes the bulk of expected losses only after the contraction arrives, which aggravates lending procyclicality and can worsen stability. Whether higher provisions reduce procyclicality depends on the cyclicity of the requirement itself and on the tax treatment of provisions. Effects are quantitatively significant in calibration. The title is the paper's verdict: EL does not by itself cure the "too little, too late" problem that motivated it.

Abad, Ikeda & Suarez (CEMFI WP 2025) — dynamic banking model with ratings migration. Extends Abad & Suarez (2018) into a calibrated dynamic model with a recursive ratings-migration structure for loan quality. Quantifies the increase in lending's sensitivity to economic conditions after the switch to expected loss and evaluates Basel III's countercyclical capital buffer as an offset, finding non-trivial trade-offs rather than a free lunch. (Full quantitative results could not be retrieved in this environment; read the paper for the headline elasticities.)

Huber (working paper, 2022; R&R at JAR per author's site) — estimated model. Develops and estimates a model of lending under accounting-based capital requirements, then counterfactually imposes CECL. CECL raises allowances relative to IL by about 16% on average and reduces lending by a few percent; magnitudes differ across circulating versions, and the paper is not yet published (do not confuse it with Yang, TAR 2025).

Buesa, Población & Tarancón (JFSR 2023) — IFRS 9 versus CECL procyclicality. Simulation on a corporate loan portfolio: IFRS 9 is less procyclical than IAS 39 but more procyclical than CECL, because IFRS 9's one-year horizon for stage 1 loans creates a cliff when loans migrate to lifetime provisioning in stage 2,

whereas CECL provisions lifetime losses from day one. The length and shape of the cycle matter more under IFRS 9. Consistent with Abad & Suarez (2018), who reached the same ranking.

Kund & Rugilo (ECB WP 2023) — stress-test simulation. Under EBA stress scenarios, IFRS 9's front-loading of losses hurts profitability and capital generation in the short run but removes the IAS 39 cliff effect, so resilience is higher in the long run and especially at the onset of a crisis.

Key findings across the theory literature

1. **CECL is not a stand-alone improvement; it is half of a policy pair.** Every analytical paper finds that the welfare effect of moving to expected loss depends on how capital regulation responds. With unchanged capital rules, EL can reduce efficiency for moderately capitalized banks when intervention is costly (Mahieux et al.). With capital re-optimized to the new information, EL lets regulators set looser requirements and expand lending (Bertomeu et al.). The U.S. five-year transition relief is a phase-in, not the re-optimization these models call for.
2. **Timeliness buys earlier intervention at the cost of precision and ex ante distortion.** The benefit of EL is ex post: earlier corrective action. The costs are ex ante: noisier reports and altered origination incentives whose sign is ambiguous. Which dominates is an empirical question about capitalization, intervention cost, and forecast precision, which is why the empirical procyclicality results are heterogeneous by bank capital.
3. **Procyclicality under EL is a statement about forecastability, not about the standard.** With rational, informative forecasts, EL smooths losses into the expansion. If downturns are unanticipated (Goncharenko & Rauf) or forecasts overreact to recent news (Vidinova), lifetime provisions are booked into the contraction and EL amplifies the cycle. The 2020 experience, where provisions spiked in Q1–Q2 and reversed in 2021, is the case these models predict.
4. **Optimal capital under CECL is state-contingent.** Vidinova, Lu & Zhang, and Abad, Ikeda & Suarez all derive time-varying capital as the complement to forward-looking provisions, with the direction depending on cycle persistence and on whether bank beliefs are rational. This connects the CECL literature to the countercyclical-buffer literature and to the Fed's open decision on incorporating CECL into the stress capital buffer.
5. **Recognition design matters: staging versus day-one lifetime loss.** IFRS 9's staging is more procyclical than CECL in simulations (Buesa et al.; Abad & Suarez) because of the stage-2 cliff, but Fleer shows a bank-specific recognition threshold can be optimal when loss reports affect depositor behavior. The trade-off between a smoother aggregate profile (CECL) and a threshold tailored to the bank's liquidity role (IFRS 9) has not been resolved in a single model.
6. **Risk-taking effects are ambiguous in sign.** Contrary to the intuition that forward-looking provisioning always disciplines lending, EL can induce riskier origination when the bank anticipates a lenient or costly intervention regime (Mahieux et al.). Evergreening incentives, by contrast, fall unambiguously under EL because rolling over a distressed loan no longer delays recognition (the mechanism tested by Eufinger et al. 2026).

What theory has not yet modelled

- **Managerial discretion and overlays.** The models treat the EL report as a noisy but unbiased signal. None models the bank choosing the bias (qualitative factors, macro-scenario weights), which is exactly what the European evidence (Behn & Couaillier 2026) says happens.
- **Bank runs and disclosure of expected losses.** Fleer is the only paper with a liquidity-provision channel; a full global-games treatment of lifetime-loss disclosure and depositor runs is missing.

- **Competition and market structure.** All models have a single bank; the empirical finding that CECL costs fall on opaque, bank-dependent borrowers suggests a lending-market model is needed.
 - **Heterogeneous banks and adoption timing.** No model explains why the 2020 adoption choice was endogenous to pandemic exposure (Aaron et al. 2025).
 - **Stress tests.** No theory paper models CECL allowances inside a supervisory stress test with a stress capital buffer, although this is the live U.S. policy question.
-

The Regulator's Perspective on the Switch to Expected-Loss Accounting

Bank regulators are not neutral observers of CECL and IFRS 9. They asked for forward-looking provisioning after the 2008 crisis, they write the capital rules that turn an accounting allowance into a binding constraint, and they have spent 2016–2026 managing the side effects of the change they requested. Their perspective has three layers: the prudential case for the standard, the policy tools they used to cushion it, and the supervisory concerns that replaced the original debate once the standard was live.

1. Why prudential regulators wanted expected-loss accounting

The incurred-loss model's "too little, too late" provisioning in 2007–09 is the origin story that every regulatory document repeats. Post-crisis, the G20, the Financial Stability Board and bank supervisors asked standard setters for provisioning that used forward-looking information; FASB's June 2016 standard and the IASB's IFRS 9 were the responses. The Basel Committee's March 2017 statement is the clearest expression of the prudential view: it "supports the use of ECL accounting approaches and encourages their application in a manner that will achieve earlier recognition of credit losses than with incurred loss models while also providing incentives for banks to follow sound credit risk management practices" (language first set out in the Committee's 2016 discussion paper). The IMF's 2026 technical note for supervisors (Awad and Riedweg) says the same in retrospect: earlier loss recognition has long had supervisors' support because it reduces the overstatement of capital in good times and strengthens loss-absorbing capacity. The U.S. agencies' June 17, 2016 joint statement took a facilitating stance, stressing that the standard applies to institutions of every size, allows multiple estimation methods, and is scalable.

2. The U.S. banking agencies: cushion the capital effect, insulate the stress test, monitor

The Federal Reserve, FDIC, OCC and NCUA have consistently separated the accounting change from its regulatory-capital consequences rather than re-designing capital around the new information.

- **Capital phase-in (2019).** The February 2019 final rule let banks phase in the day-one CET1 hit over three years, adding back 75, 50 and 25 percent of the transitional amount in years one, two and three.
- **Pandemic relief (2020).** The March 2020 interim final rule (finalized August 2020) allowed 2020 adopters to defer CECL's estimated capital effect for two years, then phase it in over three, a five-year path in total. The stated rationale was to let banks "better focus on supporting lending to creditworthy households and businesses" while "maintaining the quality of regulatory capital." The agencies also issued the May 2020 Interagency Policy Statement on Allowances for Credit Losses, which sets supervisory expectations for CECL processes, documentation, validation and board oversight.
- **Stress-test insulation (2018–2025).** In December 2018 the Fed announced it would not change its CCAR allowance framework through 2021 in order to "gather additional information on the impact of CECL." In December 2021 it extended that stance through the 2023 cycle, explaining that because the supervisory

projection starts from a firm's actual beginning-of-horizon allowance, the framework "will continue to largely offset the impact" of CECL; the aggregate CECL transition add-back in that cycle was reported as about \$28 billion, or 0.3 percent of risk-weighted assets (figure from the Fed's June 2021 stress-test results; verify against the source before quoting). In October 2024 and again in the November 2025 stress-test transparency proposal, the Board stated it is "not proposing to implement CECL into the supervisory stress testing framework," noting that its allowance model already projects provisions four quarters ahead. The result is that the stress capital buffer, the binding capital constraint for large banks, is still computed on a non-CECL allowance path.

- **Monitoring and research.** The Fed's own post-mortem (Loudis et al., FEDS Notes 2021) reported that adopters' allowances jumped 37 percent on day one and 76 percent in the first half of 2020, that provisioning was more responsive to the outlook, and that there was "limited evidence" that the allowance impact was associated with decreased lending: only the "other consumer" loan category showed a statistically significant relative decline, and the note judged even that hard to attribute to CECL given pre-existing trends. The industry read the same episode differently: the Bank Policy Institute's study of the COVID event found consumer loan growth about 3 percentage points lower at CECL banks in 2020, driven by credit card and other consumer loans, which is the source of the "3 pp" figure often cited in the policy debate. Federal Reserve economists' subsequent work is broadly favorable on information benefits (Kim, Kim, Kleymenova & Li, FEDS 2023) and cautious on model risk (Canals-Cerdá, Philadelphia Fed 2024/2026). The Atlanta Fed's 2019 review characterized the pre-adoption procyclicality evidence as mixed and model-dependent. The Kansas City Fed's August 2023 bulletin found a modest 3.76 percent average allowance increase at community banks on 2023 adoption, with material heterogeneity by size.

Within the agencies, the deposit insurer and the credit-union regulator were the skeptics for small institutions. On March 19, 2020, FDIC Chairman Jelena McWilliams wrote to FASB asking it to let current adopters postpone CECL and to impose a moratorium for institutions not yet subject to it. On April 30, 2020, NCUA Chairman Rodney Hood asked FASB to exempt credit unions outright, arguing that "the compliance costs associated with implementing CECL overwhelmingly exceed the benefits." Neither request was granted, but FASB deferred the small-entity effective date to 2023, and the NCUA's July 2021 rule phased the day-one net-worth effect in over three years and exempted credit unions under \$10 million from GAAP allowance accounting.

3. Treasury and Congress: delay, study, and "not feasible to assess"

Congress's concern was credit availability and community-bank burden. H.R. 3182 (2019), a bipartisan bill to delay CECL pending a consumer-impact study, died in committee. Section 4014 of the CARES Act (March 2020) then gave insured depository institutions, bank holding companies and their affiliates an optional delay, later extended by the Consolidated Appropriations Act, 2021 to the earlier of the first day of the fiscal year beginning after the national emergency ended or January 1, 2022. Separately, the Consolidated Appropriations Act, 2020 (December 2019) directed Treasury, in consultation with the banking agencies, to study whether CECL necessitated changes to regulatory capital requirements. Treasury's September 15, 2020 report concluded that "a definitive assessment of the impact of CECL on financial institutions' regulatory capital is not feasible at this time" and recommended that regulators continue to monitor CECL's effects on regulatory capital and lending and calibrate capital requirements as necessary (paraphrase; the exact wording could not be confirmed against the Treasury PDF in this environment). That recommendation, to recalibrate capital in light of CECL, has not been acted on beyond the transition provisions.

4. International regulators: same support, same cushioning, sharper follow-up

- **Basel Committee.** In March 2017 the Committee retained the existing regulatory treatment of provisions as an “interim” approach while it considered a longer-term treatment, and permitted jurisdictions to adopt transitional arrangements. The long-term treatment was never finalized; the interim approach persists. On April 3, 2020, the Committee gave jurisdictions flexibility to reset and extend the transitional arrangements that phase in ECL’s capital impact, while reiterating that ECL frameworks “are not designed to be applied mechanistically.”
- **European Union.** The CRR “quick fix” (Regulation (EU) 2020/873, June 2020) let banks add back 100 percent of new stage-1 and stage-2 provisions recognized in 2020–21, phasing out linearly by 2025. The ESRB’s 2017 and 2019 reports supported ECL for financial stability “if soundly implemented” but flagged procyclicality from stage transfers, over-weighting of the baseline macro scenario, and weak incentives to update forecasts. The EBA’s November 2023 monitoring report found that ECL models now recognize losses earlier but that overlays “are becoming an integral part of the ECL framework” and need tighter governance, that SICR practices diverge, and that backtesting is weak. The ECB’s May 2023 Financial Stability Review box concluded that banks with less capital headroom are at greater risk of being under-provisioned and may provision “as much as they can afford” (the within-borrower, within-period identification behind that finding is in ECB Working Paper 2841, later Behn & Couaillier 2026), and its July 2024 report on overlays found that many banks make insufficient or no additional provisions for novel risks such as geopolitical and climate risk.
- **United Kingdom.** From 2018 through the 2022/23 test the Bank of England assessed banks on an IFRS 9 transitional basis with hurdle-rate adjustments; it instructs banks to assume perfect foresight of the scenario (100 percent probability weight) when computing provisions, and explicitly recognizes that in a real stress, where no one has perfect foresight, IFRS 9 provisions could stay higher for longer. The IFRS 9 transitional arrangements expired at end-2024 and the Bank dropped hurdle-rate adjustments for the 2025 Bank Capital Stress Test.
- **IMF (2026).** The IMF technical note by Awad and Riedweg distills the supervisory playbook for jurisdictions still implementing ECL: a transition period, proportionality, keeping regulatory provisioning as a prudential backstop until supervisors have experience with IFRS 9, and parallel runs.
- **Standard setters’ reviews.** The IASB’s July 2024 post-implementation review concluded that IFRS 9’s impairment requirements are “working as intended,” with targeted disclosure improvements. FASB’s post-implementation review of CECL is ongoing; in response to private-company feedback that macroeconomic forecasting is costly and immaterial for short-term receivables, FASB issued ASU 2025-05 with practical expedients, earlier removed the troubled-debt-restructuring designation for CECL adopters (ASU 2022-02), and in November 2025 replaced purchased-credit-deteriorated accounting with a purchased-seasoned-loan model that removes the day-one “double count” (ASU 2025-08).

5. Reading the regulatory record against the academic literature

1. **Regulators wanted the standard and still defend the principle.** No prudential regulator has retreated from the case for forward-looking provisioning. The IASB’s “working as intended” verdict and the Basel Committee’s continued endorsement bracket the debate.
2. **Their response was to decouple accounting from capital, not to re-optimize capital.** Every intervention, the three-year phase-in, the pandemic add-backs in the U.S. and EU, and the Fed’s decision to keep CECL out of the stress capital buffer, insulates regulatory capital from the new allowance. This is the second-best policy in the theory literature: Bertomeu, Mahieux & Sapra (TAR 2023) and Mahieux, Sapra & Zhang (JAR 2023) show that the efficient response to better expected-loss information is to recalibrate

capital requirements jointly with the accounting change. Treasury recommended exactly that in 2020; the Basel Committee’s “interim” treatment is now nine years old.

3. **The supervisory concern has shifted from capital cliffs to discretion and model quality.** The 2016–2020 debate was about day-one capital effects and lending; the 2021–2026 supervisory output (EBA 2023, ECB 2023 and 2024, Fed and Philadelphia Fed research) is about overlays, staging judgments, backtesting, and novel risks. This tracks the academic finding that discretion survives the switch (Behn & Couaillier 2026; Novotny-Farkas et al. 2024).
4. **The regulators’ empirical read is milder than the academic one.** The Fed’s 2021 note found limited lending effects; Treasury found the question unanswerable. The top-journal evidence published since (Chen, Dou, Ryan & Zou 2025; Yang 2025; Granja & Nagel 2025; Ertan 2025) finds contraction concentrated in low-capital banks and opaque borrowers. Part of the gap is timing, since the academic papers use longer panels and loan-level data, and part is that the pandemic relief the regulators themselves granted muted the effect they were measuring.
5. **The regulatory community is not monolithic.** Prudential supervisors of large banks (Fed, OCC, Basel, ECB) are supportive; the FDIC and NCUA leadership in 2020 sought delay or exemption for small institutions; FASB has since eased the standard for private companies. The dividing line is institution size, which matches the academic evidence that CECL’s costs fall on small, information-constrained lenders and borrowers.

Gaps and Opportunities

1. **The 2023 adoption wave is under-studied.** Roughly all remaining U.S. banks adopted on 1 Jan 2023 in a non-crisis, rising-rate environment. Only descriptive work (KC Fed 2023; He et al. 2025) exists. A DiD around 2023 with 2020 adopters as controls would separate CECL’s effect from COVID and from the CARES selection problem.
 2. **Distributional effects.** Granja & Nagel (2025), Ertan (2025) and the 2026 mortgage preprint suggest costs fall on disadvantaged or opaque borrowers. Fair-lending implications of provision-driven allocation are essentially untested with rigorous identification.
 3. **Model overlays and “reasonable and supportable” forecasts.** No published U.S. paper measures the size, timing or capital-sensitivity of qualitative adjustments under CECL, although disclosure text (Kim et al.) and supervisory data (Bhojraj et al.) make it feasible.
 4. **CECL × capital regulation.** Mahieux et al. predict optimal capital should fall when early loss estimates are precise; the five-year transition ended for 2020 adopters in 2025. Whether the end of transition relief changed lending is an open, well-identified question.
 5. **Reconciling procyclicality results.** Papers use different lending margins (loan growth vs. approval sensitivity vs. pricing) and horizons. A unified loan-level study across margins, or a structural model estimated on the post-2020 panel (extending Huber 2022), would help.
 6. **Securities and non-loan assets.** Almost all work is about loans; CECL also applies to HTM securities and off-balance-sheet commitments (relevant after the 2023 regional-bank stress).
 7. **Nonbank and credit-union adoption.** Credit unions adopted in 2023 with the NCUA simplified tool; no academic evidence yet.
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Suggested Next Steps

- Read first: Mahieux, Sapra & Zhang (2023); Chen, Dou, Ryan & Zou (2025); Kim, Kim, Kleymenova & Li (2026); Granja & Nagel (2025); Aaron et al. (2025); Behn & Couaillier (2026).
 - Working papers to track for revisions: Bhojraj et al. (JAE forthcoming); Bonsall, Schmidt & Xie (2025); Gee et al. (2022, revised); Wheeler (2025, “Assessing CECL”); Eufinger et al. (2026); Bischof, Haselmann, Kohl & Schlueter (2026, TRR 266 WP 251; earlier LawFin WP 48, on internal-risk-model bias under forward-looking recognition).
 - Data: Call Reports (FFIEC 031/041, CECL indicator RCON JJ... fields), FR Y-9C, HMDA + Philadelphia Fed lender file, CRA small-business data, FR Y-14 (restricted), AnaCredit (Europe, restricted).
 - Regulatory background: Federal Reserve FEDS Notes (Dec 2021); Kansas City Fed Community Banking Bulletin (Aug 2023); Federal Register Nov 2025 stress-test transparency NPR (CECL not incorporated); IASB Post-Implementation Review of IFRS 9 Impairment (2023) and its academic literature review (Feb 2024).
 - Method: for any new 2020-adopter DiD, model adoption selection (Aaron et al.) or use approval-date designs (Yang).
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Post-Flight Verification

Round 3 (regulator’s perspective, 28 claims): PARTIAL — 25 supported; 2 contradicted and corrected (a Bank Policy Institute lending result had been attributed to the Fed’s FEDS Note; the Treasury study was mandated by the Consolidated Appropriations Act, 2020, not CARES Act §4014); 4 low-severity precision notes applied (2019 phase-in mechanics, Basel vs. EU add-back scope, Bank of England 2025 change, FASB ASU 2025-08). Two figures remain unverified against blocked source PDFs and are flagged inline: the Fed’s \$28 billion / 0.3 percent transition amount and the exact wording of Treasury’s recommendation.

Round 2 (theory deep-dive, 10 claims): PASS — all ten theory papers located at the stated venues with matching volume, pages, DOI or working-paper number, and all reported findings supported by abstract text. Three low-severity notes applied: Lu & Zhang’s comparative-statics wording now follows the published version; Huber’s “larger gap in downturns” sub-claim dropped as unconfirmed; JFSR issue number and CEMFI paper number added. The verifier also confirmed Huber’s paper is unpublished (R&R at JAR), so the 16% figure is cited to the SSRN working paper, not to a journal.

Round 1 (43 claims): PARTIAL — 36 of 43 claims SUPPORTED as written; 3 corrected (C1 Mahieux et al. finding overstated, C8 author count, C23 journal year); 3 magnitude/detail claims softened (Huber version conflict, Jia et al. magnitudes, Yang loan-sale and Chen-Lin-Yang housing-boom secondary results); 2 author lists unresolved (JAPP 2025 management forecasts; Research Square 2026 mortgage preprint). **No fabricated citations found:** every paper, working paper, preprint and regulatory note was located at the named venue.

▼ Verifier method and per-claim outcome

Verification ran in a forked `claim-verifier` context that received only the 43-row claims table, never the draft. Publisher, SSRN, Fed and API endpoints (Crossref, OpenAlex, Semantic Scholar) are egress-blocked in this environment, so evidence rests on search-result snippets quoting the source pages; anything not corroborated by such a snippet was marked unverifiable rather than guessed. | Outcome | Claims | |---|---| | Supported, cited as written | Kim et al. 2026; Chen, Dou, Ryan & Zou 2025; Granja & Nagel 2025; Ertan 2025; Behn &

Couaillier 2026; Lu & Nikolaev 2022; Bhojraj et al.; Beatty & Liao 2021; Wheeler 2021; López-Espinosa et al. 2021; Kim 2022; Loudis et al. 2021 (all four figures); Canals-Cerdá 2024/2026; Novotny-Farkas et al. 2024; Buchetti et al. 2025; Li, Ng & Saffar 2024; Bonsall et al. 2025; Bonaldi et al. 2023; Wheeler 2025; Vidinova 2023; Eufinger et al. 2026; Guo et al. 2025; Kund & Rugilo 2023; KC Fed 2023 (all three figures); Hu 2024b; Federal Register 2025 stress-test NPR || Corrected in this version | Mahieux et al. 2023 (finding is conditional; origination can be safer or riskier); Chen & Huang (two authors, 2026); Elashmawy & Kallunki (Accounting in Europe 23(1) is 2026); titles of Yang 2025, Qiang & Wang 2024, Hu 2024a, Bable et al. 2023; Aaron et al. full citation 42(4): 2915–2948 || Softened / flagged | Huber 2022 magnitudes (two versions circulate); Jia et al. 2025 magnitudes (abstract-level claims only); Yang 2025 loan-sale result; Chen, Lin & Yang 2023 housing-boom result; Fleer 2025 article number; Gee et al. revision date || Unresolved | Authors of JAPP 2025 "Expected loss recognition and banks' management forecasts"; author(s) of Research Square rs-10696408/v1 |

Appendix: BibTeX Entries

Entries marked `VERIFY` carry a detail that could not be confirmed against the publisher page; resolve with library access before citing.

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@article{LiNgSaffar2024,
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  title = {Accounting-Driven Bank Monitoring and Firms' Debt Structure: Evidence from {IFRS} 9 Adoption},
  journal = {Management Science},
  year = {2024},
  volume = {70},
  number = {1},
  pages = {54--77},
  doi = {10.1287/mnsc.2022.4628}
}

@article{Fleer2025,
  author = {Fleer, Sebastian},
  title = {Significant Expected Lifetime Credit Loss Impairments: Determinants of Bank Loss Recognition and Stability Implications},
  journal = {Journal of Accounting and Public Policy},
  year = {2025},
  volume = {51},
  note = {VERIFY: article number (107313 unconfirmed); author attribution from the SSRN version (4131096). PII S0278425425000249}
}

@article{JAPP2025ManagementForecasts,
  author = {[authors not identified in search results]},
  title = {Expected Loss Recognition and Banks' Management Forecasts},
  journal = {Journal of Accounting and Public Policy},
  year = {2025},
  note = {VERIFY: authors, volume, article number. ScienceDirect PII S0278425425000882}
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@article{BertomeuMahieuxSapra2023,
  author = {Bertomeu, Jeremy and Mahieux, Lucas and Sapra, Haresh},
  title = {Interplay between Accounting and Prudential Regulation},
  journal = {The Accounting Review},
  year = {2023},
  volume = {98},
  number = {1},
  pages = {29--53},
  note = {SSRN 3266348}
}

@article{GoncharenkoRauf2024,
  author = {Goncharenko, Roman and Rauf, Asad},
  title = {Still ``Too Much, Too Late``: Provisioning for Expected Loan Losses},
  journal = {International Journal of Central Banking},
  year = {2024},
  volume = {20},
  number = {4},
  pages = {415--474},
  note = {SSRN 3247631. Earlier title: ``Loan Loss Provisioning Requirements in a Dynamic Model of Banking''}
}

@article{BuesaPoblacionTarancon2023,
  author = {Buesa, Alejandro and Poblacion, Javier and Tarancon, Javier},
  title = {The Procyclicality of Impairment Accounting: Comparing Expected Losses under IFRS 9 and US GAAP},
  journal = {Journal of Financial Services Research},
  year = {2023},
  volume = {64},
  number = {3},
  pages = {303--324},
  doi = {10.1007/s10693-022-00392-1}
}

@article{LuZhang2026,
  author = {Lu, Tong and Zhang, Lanyi Yan},
  title = {Time-Varying Capital Ratios under CECL},
  journal = {Advances in Accounting},
  year = {2026},
  volume = {70},
  pages = {100865},
  note = {SSRN 4710219 (working-paper version, Jan 2024); ScienceDirect PII S0882611025000604}
}

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@unpublished{BhojrajKleyменоваLiuLuSengupta2025,
  author = {Bhojraj, Sanjeev and Kleyменова, Anya and Liu, Miao and Lu, Yao and Sengupta, Partha},
  title = {Imperfect Expectations in Loan Loss Forecasts},
  year = {2025},
  note = {SSRN 4858012; reported as accepted at Journal of Accounting and Economics. VERIFY status}
}

@unpublished{GeeNeilsonSchmidtXie2025,
  author = {Gee, Kurt H. and Neilson, Jed and Schmidt, Brent and Xie, Biqin},
  title = {When Are Expected Credit Losses Decision-Useful and New to Investors? Evidence from CECL Adoption},
  year = {2025},
  note = {SSRN 4038479, first posted Feb 2022; revision date unconfirmed}
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@unpublished{BonsallSchmidtXie2025,
  author = {Bonsall, Samuel B. and Schmidt, Brent and Xie, Biqin},
  title = {Current Expected Credit Loss ({CECL}) Model and Banks' External Information Environment},
  year = {2025},
  note = {SSRN 4236547, revised May 2025. Earlier related: ``CECL Model and Analyst Forecasts'' (2022)}
}

@unpublished{BonalDiLiangYang2023,
  author = {BonalDi, Pietro and Liang, Pierre Jinghong and Yang, Lavender},
  title = {Did {CECL} Improve Banks' Loan Loss Provisions and Earnings Quality during the {COVID-19} Pandemic?},
  year = {2023},
  note = {SSRN 4360356}
}

@unpublished{ChenLinYang2023,
  author = {Chen, Yi-Chun and Lin, Tse-Chun and Yang, Donghe},
  title = {Current Expected Credit Loss Standard and Bank Lending Procyclicality: Evidence from the {US} Housing Market},
  year = {2023},
  note = {SSRN 4410897}
}

@unpublished{Hu2024SmallBusiness,
  author = {Hu, Xiaoyu},
  title = {The Real Effects of Current Expected Credit Loss Provision ({CECL}) on Banks' Small Business Lending},
  year = {2024},
  note = {SSRN 4934238}
}

@unpublished{Hu2024Spillover,
  author = {Hu, Xiaoyu},
  title = {The Impact of Current Expected Credit Loss Provision ({CECL}) on the Information Transmission of Banks' Loan Loss Provision},
  year = {2024},
  note = {SSRN 4782147}
}

@unpublished{Wheeler2025,
  author = {Wheeler, P. Barrett},
  title = {Assessing {CECL}: An Examination of the Informativeness of Loan Loss Provisions Before and After Adoption},
  year = {2025},
  note = {SSRN 5400903}
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@unpublished{BableWongWynes2023,
  author = {Bable, Jordan and Wong, Christopher and Wynes, Michael J.},
  title = {The Decision Usefulness of {CECL}: Users' Views about the Current Expected Credit Losses Model},
  year = {2023},
  note = {SSRN 4483117}
}

@unpublished{Huber2022,
  author = {Huber, Stefan},
  title = {Loan Loss Measurement and Bank Lending},
  year = {2022},
  note = {SSRN 4120628. R\&R at Journal of Accounting Research per author's website.

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Two versions circulate with different magnitudes (14.7\%/5.16\% vs 16\%/3.15\%); cite the dated version}
}

@unpublished{Vidinova2023,
 author = {Vidinova, Hristiana},
 title = {Forward-Looking Loan Loss Provisioning under Imperfect Forecasts},
 year = {2023},
 note = {SSRN 4628656}
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 author = {Novotny-Farkas, Zolt{\`a}n and Oberson, Romain and Renner, Elisabeth},
 title = {{IFRS} 9 under Stress: Loan Loss Provisioning under the Expected Credit Loss Model},
 year = {2024},
 note = {SSRN 4918921}
}

@unpublished{EufingerRamirezChiangSakasai2026,
 author = {Eufinger, Christian and Ramirez-Chiang, Guillermo and Sakasai, Yuki},
 title = {{(N)}evergreening Loans: From Incurred to Expected Credit Loss Models},
 year = {2026},
 note = {SSRN 7216299}
}

@unpublished{BischofHaselmannKohlSchlueter2026,
 author = {Bischof, Jannis and Haselmann, Rainer F. H. and Kohl, Frederik and Schlueter, Oliver},
 title = {Forward-Looking Credit Loss Recognition and Banks' Internal Risk Models: Timeliness, Reporting Bias, and Lending Effects},
 year = {2026},
 note = {SSRN 7386038 (Sept 2026); TRR 266 WP 251. An earlier 2022 version circulated as LawFin WP 48 under the title ``Limitations of implementing an expected credit loss model''}
}

@unpublished{GuoJiaNgZhu2025,
 author = {Guo, Jia and Jia, Yifan and Ng, Jeffrey and Zhu, Haoran},
 title = {Spillover Effects of Timelier Loan Loss Recognition by Banks on Borrowers' Cross-Border Borrowing},
 year = {2025},
 note = {SSRN 4958491, posted May 2025}
}

@techreport{AbadIkedaSuarez2025,
 author = {Abad, Jorge and Ikeda, Daisuke and Suarez, Javier},
 title = {From Incurred to Expected Loss: Implications for Bank Lending},
 institution = {CEMFI},
 type = {Working Paper},
 number = {2509},
 year = {2025},
 month = may,
 note = {RePEc handle cmf/wpaper/wp2025_2509}
}

@techreport{CanalsCerdà2024,
 author = {Canals-Cerd{\`a}, Jos{\`e} J.},
 title = {{CECL} Implementation and Model Risk in Uncertain Times: An Application to Consumer Finance},
 institution = {Federal Reserve Bank of Philadelphia},
 type = {Working Paper},
 number = {24-03},
 year = {2024},

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    doi      = {10.21799/frbp.wp.2024.03},
    note     = {Superseded by WP 26-09}
}

@techreport{CanalsCerde2026,
  author    = {Canals-Cerde{\`a}, Jos{\`e} J.},
  title     = {Model Risk Under {CECL}: A Consumer Finance Perspective},
  institution = {Federal Reserve Bank of Philadelphia},
  type      = {Working Paper},
  number    = {26-09},
  year      = {2026}
}

@techreport{KundRugilo2023,
  author    = {Kund, Arndt-Gerit and Rugilo, Daniel},
  title     = {Does {IFRS} 9 Increase Banks' Resilience?},
  institution = {European Central Bank},
  type      = {Working Paper},
  number    = {2792},
  year      = {2023}
}

@misc{LoudisEtAl2021,
  author    = {Loudis, Bert and Pechenik, Sasha and Ranish, Ben and Vojtech, Cindy M.
and Xu, Helen},
  title     = {New Accounting Framework Faces Its First Test: {CECL} During the
Pandemic},
  howpublished = {FEDS Notes, Board of Governors of the Federal Reserve System},
  month     = dec,
  year      = {2021},
  doi       = {10.17016/2380-7172.3025}
}

@misc{KansasCityFed2023,
  author    = {{Federal Reserve Bank of Kansas City}},
  title     = {Highlight: {CECL} Adoption's Impact on Community Bank Allowance
Levels},
  howpublished = {Community Banking Bulletin},
  month     = aug,
  year      = {2023}
}

@misc{ResearchSquare2026Mortgage,
  author    = {[author not identified in search results]},
  title     = {Expected-Loss Provisioning and the Distribution of Mortgage Credit},
  howpublished = {Research Square preprint rs-10696408/v1},
  year      = {2026},
  note     = {VERIFY: author, date; preliminary, not peer reviewed}
}

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@misc{AgenciesJointStatement2016,
  author    = {{Board of Governors of the Federal Reserve System and FDIC and NCUA
and OCC}},
  title     = {Joint Statement on the New Accounting Standard on Financial
Instruments -- Credit Losses},
  howpublished = {Interagency statement},
  month     = jun,
  day       = {17},
  year      = {2016}
}

@misc{AgenciesCECLCapitalRule2019,
  author    = {{OCC and Board of Governors of the Federal Reserve System and FDIC}},

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    title      = {Regulatory Capital Rule: Implementation and Transition of the Current
Expected Credit Losses Methodology for Allowances},
    howpublished = {Federal Register, final rule},
    month      = feb,
    day        = {14},
    year       = {2019}
}

@misc{AgenciesCECLTransitionIFR2020,
    author      = {{OCC and Board of Governors of the Federal Reserve System and FDIC}},
    title       = {Regulatory Capital Rule: Revised Transition of the Current Expected
Credit Losses Methodology for Allowances},
    howpublished = {Interim final rule (March 2020); final rule, Federal Register 85 FR
61577},
    month      = sep,
    day        = {30},
    year       = {2020}
}

@misc{InteragencyACLPolicy2020,
    author      = {{Board of Governors of the Federal Reserve System and FDIC and NCUA
and OCC}},
    title       = {Interagency Policy Statement on Allowances for Credit Losses},
    howpublished = {Federal Register 85 FR 32991},
    month      = jun,
    day        = {1},
    year       = {2020}
}

@misc{FedCECLStressTest2018,
    author      = {{Board of Governors of the Federal Reserve System}},
    title       = {Statement on the Current Expected Credit Loss Methodology ({CECL}) and
Stress Testing},
    month      = dec,
    day        = {21},
    year       = {2018}
}

@misc{FedCECLStressTest2021,
    author      = {{Board of Governors of the Federal Reserve System}},
    title       = {Public Statement on the Treatment of {CECL} in the Supervisory Stress
Test},
    month      = dec,
    day        = {9},
    year       = {2021}
}

@misc{FedStressTestNPR2025,
    author      = {{Board of Governors of the Federal Reserve System}},
    title       = {Enhanced Transparency and Public Accountability of the Supervisory
Stress Test Models and Scenarios; Modifications to the Capital Planning and Stress
Capital Buffer Requirement Rule},
    howpublished = {Federal Register, notice of proposed rulemaking, 2025-20211},
    month      = nov,
    day        = {18},
    year       = {2025}
}

@misc{Treasury2020CECL,
    author      = {{U.S. Department of the Treasury}},
    title       = {The Current Expected Credit Loss Accounting Standard and Financial
Institution Regulatory Capital},
    howpublished = {Study directed by the Consolidated Appropriations Act, 2020 (S. Rept.
116-111)},

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    month      = sep,
    day        = {15},
    year       = {2020}
}

@misc{McWilliams2020FASB,
  author      = {McWilliams, Jelena},
  title       = {Letter to the Financial Accounting Standards Board on {COVID-19}
relief, troubled debt restructurings, and {CECL}},
  howpublished = {FDIC},
  month       = mar,
  day         = {19},
  year        = {2020}
}

@misc{Hood2020FASB,
  author      = {Hood, Rodney E.},
  title       = {Letter to the Financial Accounting Standards Board requesting
exemption of credit unions from {CECL}},
  howpublished = {NCUA},
  month       = apr,
  day         = {30},
  year        = {2020}
}

@misc{NCUACECLRule2021,
  author      = {{National Credit Union Administration}},
  title       = {Transition to the Current Expected Credit Loss Methodology},
  howpublished = {Federal Register, final rule},
  month       = jul,
  day         = {1},
  year        = {2021}
}

@misc{CRS2018CECL,
  author      = {{Congressional Research Service}},
  title       = {Banking: Current Expected Credit Loss ({CECL})},
  howpublished = {CRS Report R45339},
  month       = oct,
  year        = {2018}
}

@misc{BCBS2017Provisions,
  author      = {{Basel Committee on Banking Supervision}},
  title       = {Regulatory Treatment of Accounting Provisions -- Interim Approach and
Transitional Arrangements},
  howpublished = {BIS, d401},
  month       = mar,
  year        = {2017}
}

@misc{BCBS2020Covid,
  author      = {{Basel Committee on Banking Supervision}},
  title       = {Measures to Reflect the Impact of {Covid-19}},
  howpublished = {BIS, d498},
  month       = apr,
  day         = {3},
  year        = {2020}
}

@misc{EUCRRQuickFix2020,
  author      = {{European Parliament and Council}},
  title       = {Regulation ({EU}) 2020/873 amending Regulations ({EU}) No 575/2013 and
({EU}) 2019/876 as regards certain adjustments in response to the {COVID-19} pandemic},

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    howpublished = {Official Journal of the European Union},
    month        = jun,
    day          = {26},
    year         = {2020}
}

@misc{ESRB2019ECL,
  author      = {{European Systemic Risk Board}},
  title       = {The Cyclical Behaviour of the {ECL} Model in {IFRS} 9},
  month       = mar,
  year        = {2019}
}

@misc{EBA2023IFRS9,
  author      = {{European Banking Authority}},
  title       = {{IFRS} 9 Implementation by {EU} Institutions: Monitoring Report},
  month       = nov,
  year        = {2023}
}

@misc{ECB2024Overlays,
  author      = {{European Central Bank Banking Supervision}},
  title       = {{IFRS} 9 Overlays and Model Improvements for Novel Risks},
  month       = jul,
  year        = {2024}
}

@misc{IASB2024PIR,
  author      = {{International Accounting Standards Board}},
  title       = {Post-implementation Review of {IFRS} 9 Financial Instruments --
Impairment: Project Summary and Feedback Statement},
  month       = jul,
  year        = {2024}
}

@techreport{IMF2026IFRS9,
  author      = {Awad, Rachid and Riedweg, Laurent},
  title       = {{IFRS} 9 Implementation from the Perspective of Banking Supervisors},
  institution = {International Monetary Fund},
  type        = {Technical Notes and Manuals},
  number      = {2026/004},
  year        = {2026}
}

@misc{Wall2019AtlantaFed,
  author      = {Wall, Larry D.},
  title       = {Procyclicality: {CECL} versus Incurred Loss Model},
  howpublished = {Federal Reserve Bank of Atlanta, Notes from the Vault},
  month       = oct,
  year        = {2019}
}

@misc{FASB2025ASU05,
  author      = {{Financial Accounting Standards Board}},
  title       = {Accounting Standards Update 2025-05, Financial Instruments -- Credit
Losses (Topic 326): Measurement of Credit Losses for Accounts Receivable and Contract
Assets},
  year        = {2025}
}

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